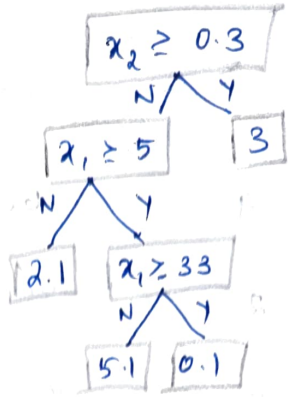
② Regression tree

x_1 : temp

x_2 : precipitation level

y : Km/kat Iran



11/02/26

Predict the risk for Autism

IQ Score	social Score	verbal Score	Risk
100	79	86	Low
30	40	70	High
80	85	65	Low
90	82	45	High

Test Sample

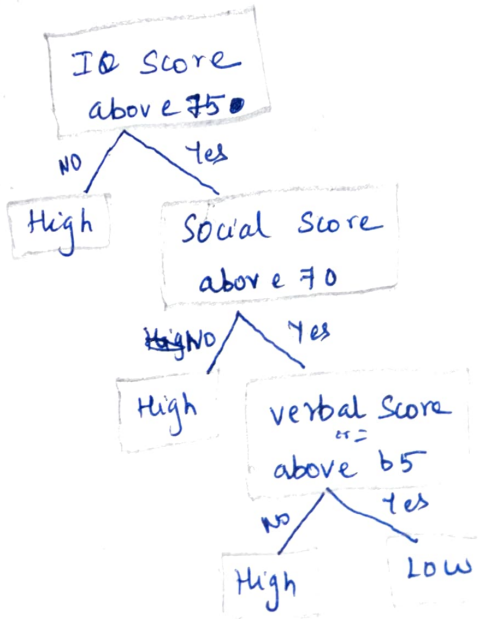
IQ = 120

social Score = 66

verbal = 40

Risk of autism

My tree not by mathematical



Area	Bedroom	Price
1700	4	60
1500	2	30
1200	3	40
1100	1	20

Parameters

Split dimension
Split value
child nodes

x_1	x_2	x_3	y
2	3	1	5
1	2	2	4
3	1	2	6
4	3	3	9
5	2	1	10
6	1	2	12

1) Take feature x_1

2) Sort feature value
[1, 2, 3, 4, 5, 6]

3) Candidate split point
[1.5, 2.5, 3.5, 4.5, 5.5, 6.5]

$x_1 \leq 3.5$

2, 1, 3	4, 5, 6
5, 4, 6	9, 10, 12

$$MSE = \frac{(5-5)^2 + (4-5)^2 + (6-5)^2}{3} = 0.67$$

$\hat{y} = 5$

$\frac{5+4+6}{3} = 5$

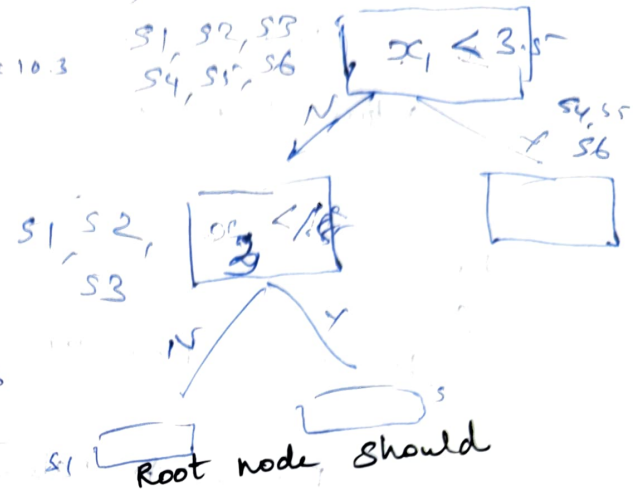
$\hat{y} = 10.3$

$\frac{9+10+12}{3} = 10.3$

$$\frac{(9-10.33)^2 + (10-10.33)^2 + (12-10.33)^2}{3} = 1.56$$

Total error: $0.67 + 1.56$

$= 2.2$



Feature	Candidate split value	Loss
x_1	1.5	l_1
	2.5	l_2
	3.5	2.2
	4.5	l_4
	5.5	l_5
x_2	c_1	l_1
	c_2	l_2
	⋮	⋮
	c_n	l_n

Root node should be lowest loss among all the feature x_1, x_2, x_3

Hyperparameter less than or ≤ 2

leaf node less than or $\leq k$ element